ASCII MASTER FLOWCHART

Earthquake Risk and Resilient Construction

12-panel revision companion | GS-III Disaster Management

ASCII MASTER FLOW - PANEL 1/12: Tectonic risk chain

```
PLATE CONVERGENCE -> FAULT STRESS -> RUPTURE -> SEISMIC WAVES
HAZARD x EXPOSURE x VULNERABILITY / CAPACITY -> RISK
OCCURRENCE UNCONTROLLABLE -> VULNERABILITY REDUCIBLE
```

ASCII MASTER FLOW - PANEL 2/12: Magnitude and intensity

```
MAGNITUDE -> source energy; commonly Mw; one event value
INTENSITY -> observed effects at a place; MMI / MSK
VARIES WITH -> distance, depth, ground and building stock
```

ASCII MASTER FLOW - PANEL 3/12: India seismic map

```
IS 1893:2016 -> ZONES II, III, IV, V
ZONE V -> whole NE + A&N + parts of northern Bihar,
          J&K/Ladakh, Himachal, Uttarakhand and Gujarat
ZONE V 11% | IV 18% | III 30%
ZONES III-V -> about 59% of landmass
MARCH 2026 REVISION WITHDRAWN -> IS 1893:2016 OPERATIVE
```

ASCII MASTER FLOW - PANEL 4/12: Site-effect route

```
REGIONAL ZONE + SOIL / BASIN / SLOPE / GROUNDWATER
-> AMPLIFICATION / LIQUEFACTION / SLOPE FAILURE
-> MICROZONATION
-> LOCAL LAND-USE AND DESIGN ACTION
```

ASCII MASTER FLOW - PANEL 5/12: Constructed vulnerability

```
WEAK MASONRY / SOFT STOREY / POOR CONNECTIONS
FALLING COMPONENTS / FIRE / BLOCKED ACCESS
LIFELINE FAILURE -> CASCADING LOSS
CASES -> LATUR | BHUJ | KASHMIR | SIKKIM
```

ASCII MASTER FLOW - PANEL 6/12: Mitigation system

```
STRUCTURAL -> frame, walls, foundation, connections, ductility
NON-STRUCTURAL -> tanks, ceilings, shelves, equipment, utilities
HOUSEHOLD -> anchor, brace, clear exits
ALL LEVELS -> INSPECT AND MAINTAIN
```

ASCII MASTER FLOW - PANEL 7/12: NDMA six pillars

```
1 SAFE NEW CONSTRUCTION
2 SELECTIVE RETROFITTING
3 REGULATION AND ENFORCEMENT
4 AWARENESS AND PREPAREDNESS
5 CAPACITY DEVELOPMENT
6 EMERGENCY RESPONSE
```

ASCII MASTER FLOW - PANEL 8/12: Compliance chain

```
ZONE / SITE -> DESIGN -> DETAIL -> MATERIAL / WORKMANSHIP
-> INSPECTION -> AS-BUILT VERIFICATION -> MAINTENANCE
CODE ON PAPER != SAFE BUILDING
```

ASCII MASTER FLOW - PANEL 9/12: Existing stock and lifelines

```
INVENTORY -> SCREEN -> AUDIT -> PRIORITISE
-> FINANCE -> RETROFIT -> VERIFY
HOSPITAL / SCHOOL / BRIDGE / EOC / UTILITY
COLLAPSE PREVENTION + FUNCTIONAL CONTINUITY
```

ASCII MASTER FLOW - PANEL 10/12: Institutions and tools

```
NERMP -> structural and non-structural mitigation
BMTPC -> disaster-resistant construction
VULNERABILITY ATLAS 2019 -> hazard + housing stock
NCS -> 170 observatories + BhooKamp (1 April 2026)
```

ASCII MASTER FLOW - PANEL 11/12: Preparedness and response

```
BEFORE -> drill, household safety, inventory, continuity plan
IMPACT -> EOC ACTIVATES IRS
IRS -> command, operations, planning and logistics
AFTER -> rescue, medical aid, damage and needs assessment
```

ASCII MASTER FLOW - PANEL 12/12: Answer spine and PYQ boundary

```
2021 DIRECT -> vulnerability + regional examples + mitigation
2026 ADJACENT -> risk-sensitive infrastructure support only
THESIS -> no precise prediction; compliance is mitigation
VERDICT -> safer stock, lifelines and accountable enforcement
```